

Professor Dimas | Guide Manual | Introduction to Power BI

Make sure you first install Power BI desktop on your computer. These instructions are designed for specifically Power BI desktop and they may vary if using the online version of Power BI.

Dataset Summary / Description

The National Student Loan Data System (NSLDS) is the national database of information about loans and grants awarded to students under Title IV of the Higher Education Act (HEA) of 1965. NSLDS provides a centralized, integrated view of Title IV loans and grants during their complete life cycle, from aid approval through disbursement, repayment, deferment, delinquency, and closure.

The dataset can be downloaded using the following link: <https://drive.google.com/file/d/1RU7L4-XoQxwPKZ02Al0whB4OnfrMGxN/view?usp=sharing>

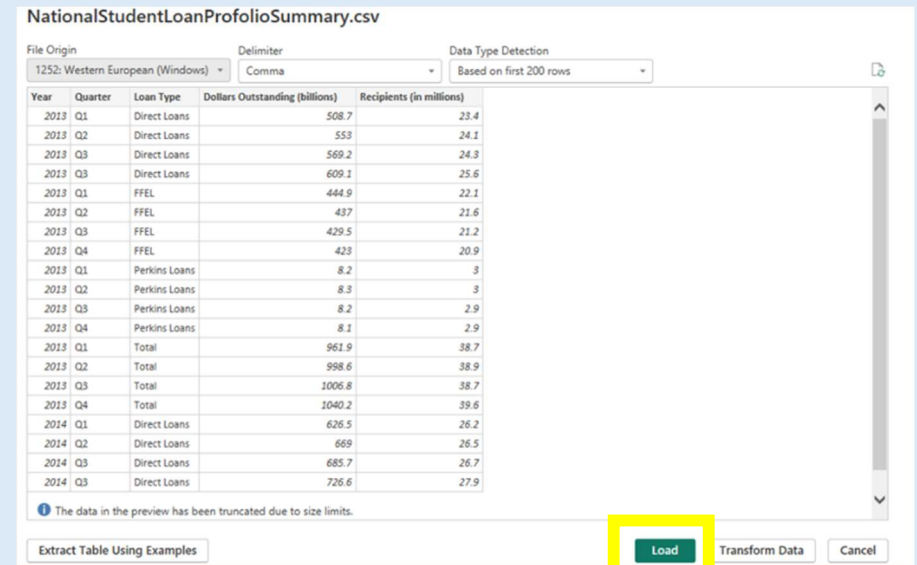
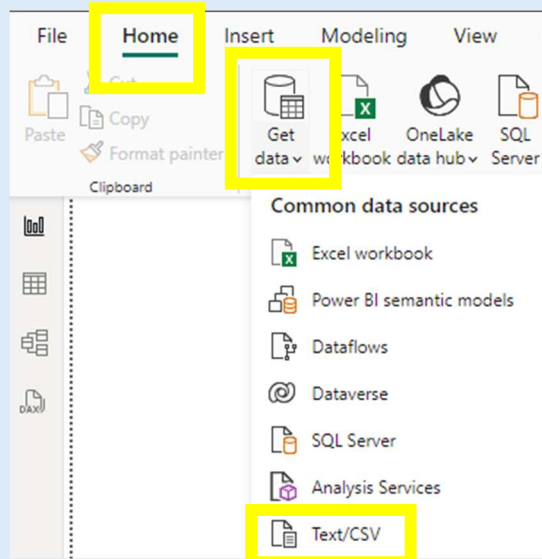
Original Source: <https://catalog.data.gov/dataset/national-student-loan-data-system-722b0> (Data Pulled in 2024 - PortfolioSummary)

Features:

Year	Loan year information
Quarter	Yearly quarters: Q1,Q2,Q3,Q4
Loan Type	Type of National Student loan: Direct loan, FFEL(Federal Family Education Loan), Perkins Loans
Dollars Outstanding (in billions)	Amount of student loan debt that is still owed and has not yet been repaid
Recipients (in millions)	Number of individuals who have received or currently hold federal student loans

1. Open Power BI and save the file. → File → Save As
2. Load the csv file “NationalStudentLoanProfolioSummary.csv” using the “**Get data**” and “**Text/CSV**” option under the home tab
 - a. Explore the options for **Delimiter and Data Type Detection**. What do they control? (Use default settings to load data)

Step 2a.



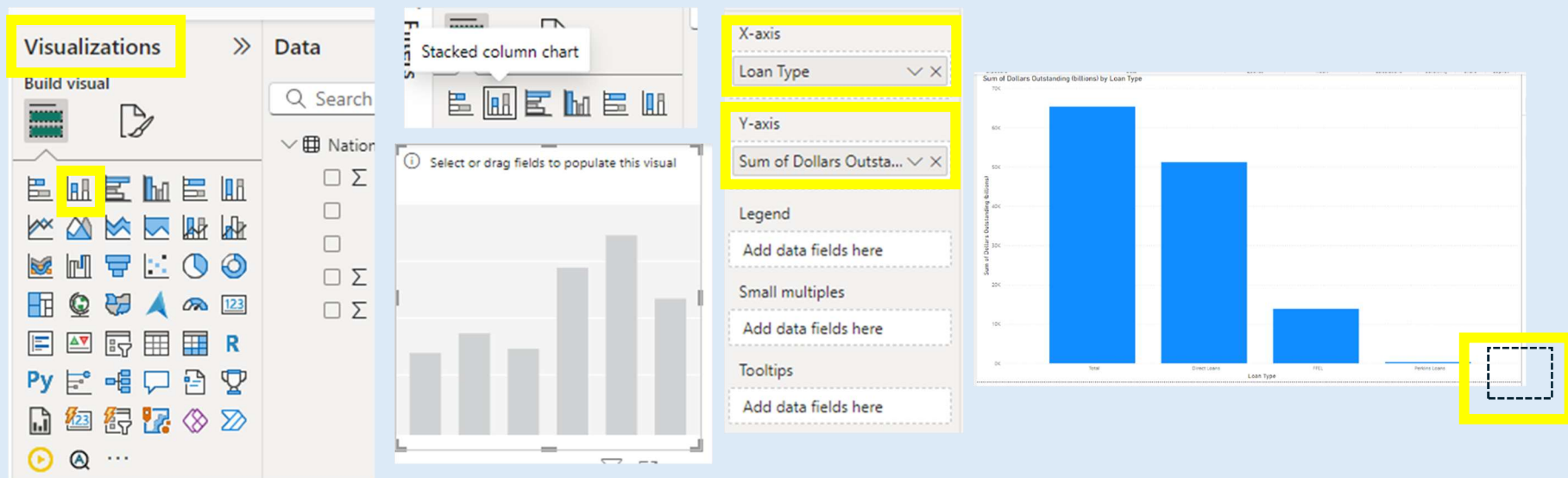
Bar Chart

3. Now let's build a bar chart. Under the **Visualizations** section, select the “**Stacked column chart**” icon
 - a. On X-axis shelf drag and drop “**Loan Type**” and on Y-axis add “**Dollars Outstanding (billions)**”
 - b. Using the lower right corner of the visual, you can drag the corner to increase or decrease the size of the visual to fit the space

Note: In Power BI you can create a single visual or multiple visuals on a single page. For this introduction guide we will create each visual on its own page.

IMPORTANT: Power BI automatically adjusts the x and y-axis value ranges dynamically when we change the size of the chart. Meaning if your final visualization is different for the ranges of x or y axis shown compared to the examples don't fret.

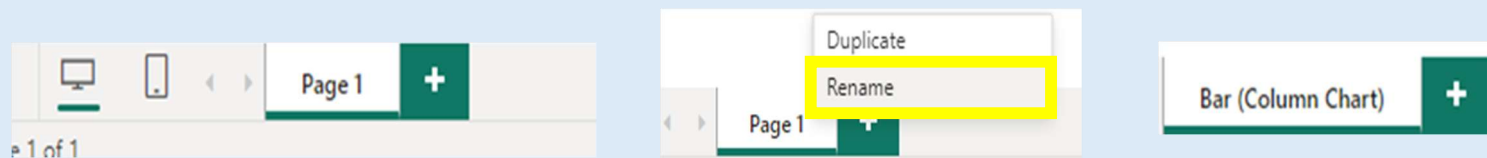
Step 3a.-3b.



4. Change the name of the visualization at the bottom tab from “**Page 1**” to “**Bar Chart (Column)**”

Note: This is optional, but is recommended if you are creating multiple pages or visuals in a single file)

Step 4

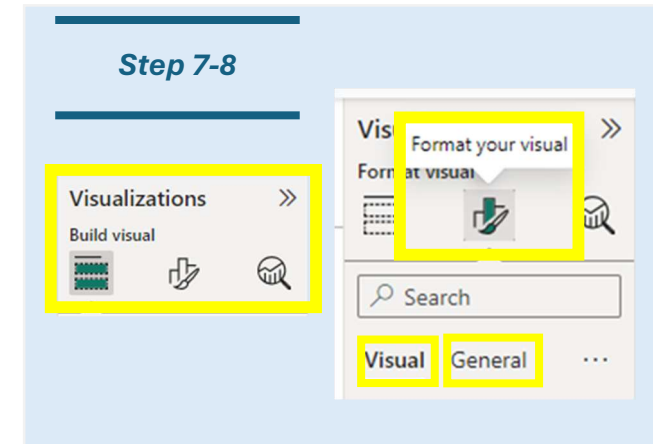


5. Test out what happens when you **choose other bar chart options** (first select the visualization you already created, then in the visualization tab you can change to other chart types by clicking on other icons.)

Note: not all charts will work with the data we have plotted

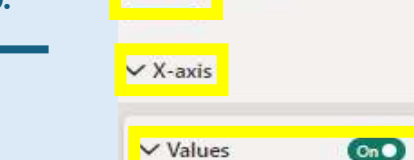
- Before moving onto the next step, return the visualization to a **“Stacked column chart”** as we did in step #3
- Now let’s make the chart “pretty”. **Select the visualization**, then under the visualization options menu (*to the right of the visual where we build the visuals there will be three boxes at the top*), change from **“Build visual”** to the **“Format your visual”**.
- Take a moment to look at the different type of options there are in **“Format your visual”** for both the **“Visual”** and **“General”** menus.

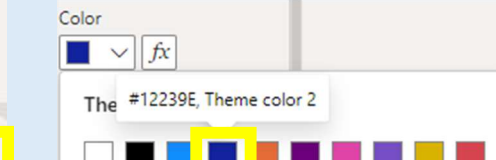
Note: “Visual” controls everything to do with the actual data plotted within the visual, and “General” controls the overall structure and whole of the visual such as visual title, border of visual, etc.



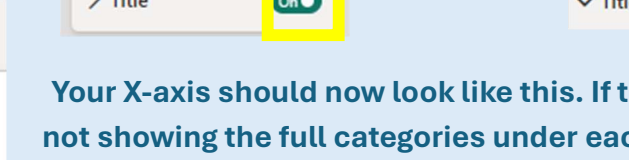
9. Let's edit and format the **X-axis** and **Y-axis** font text, Click the “**Visual**” option to access elements specific to the visualization
 - a. Under **X-axis** and **Values** change the **font text size to 15** and color to the “**#12239E, Theme color 2**” and **Bold** the text
 - b. Under **X-axis** and **Title**, turn **off the title for the x-axis**, this is redundant information we don't need

Step 9a.-9b.



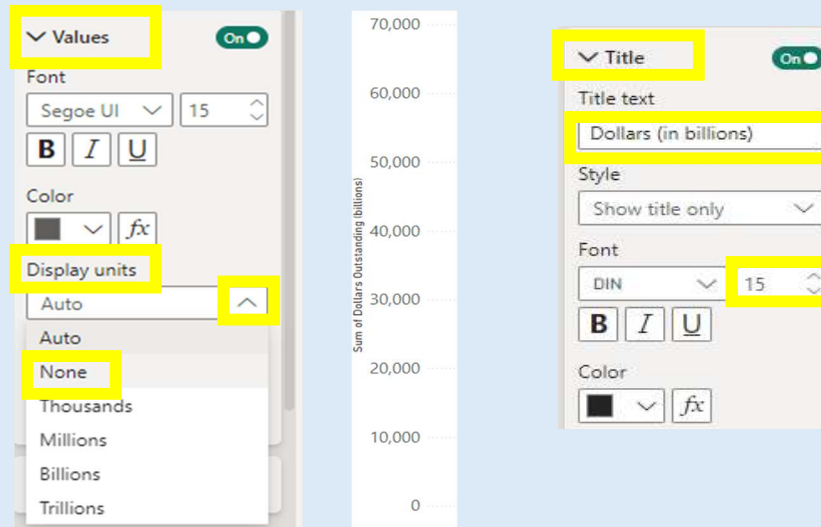


Your X-axis should now look like this. If the text is not showing the full categories under each bar you may need to increase the size of your chart



- c. Under the **Y-axis** and **Values** change the font text size to **15** and change “**Display Units**” from Auto to None
- d. Change the title of the **Y-axis** to “**Dollars (in billions)**” and increase the **font size to 15**

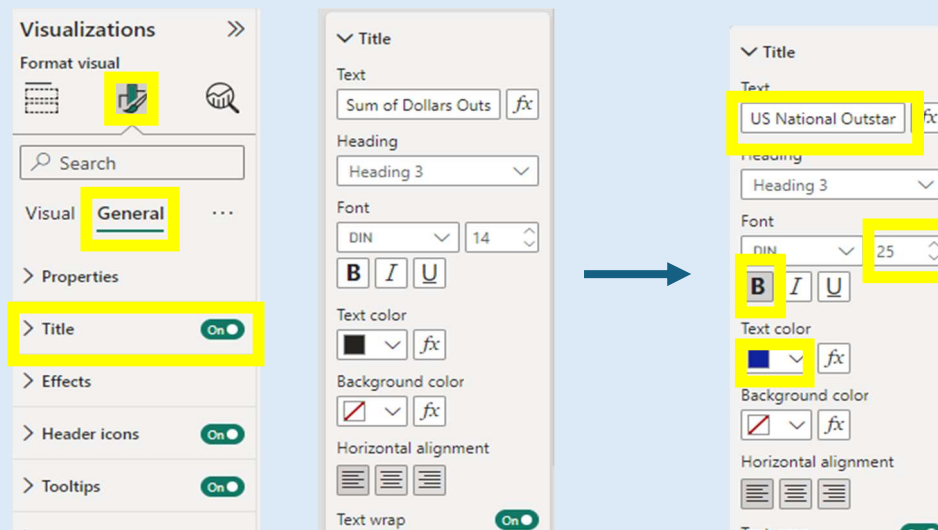
Step 9c.-9d.



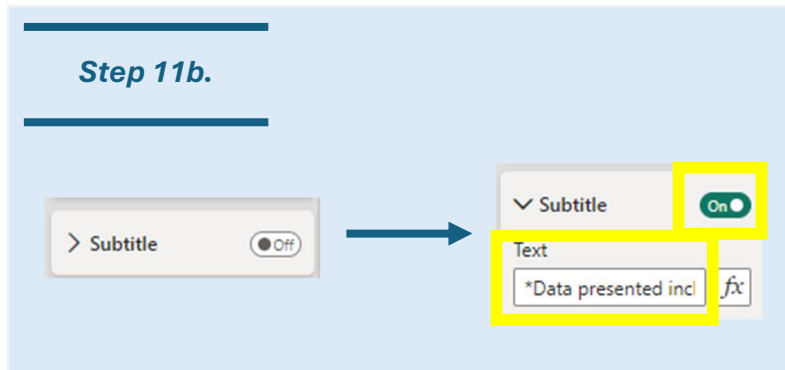
10. Let's edit/format the Title for the entire visualization

- a. Under “**Format visual**” select “**General**” then “**Title**” and change the visualization title to “**Total US National Outstanding Loan Amounts 2013-2024* by Type**”
- b. Change text color to “**#12239E, Theme color 2**” and **Bold** the text, **font size 25** (note: you may need to make the visual bigger)

Step 10a.-10b.

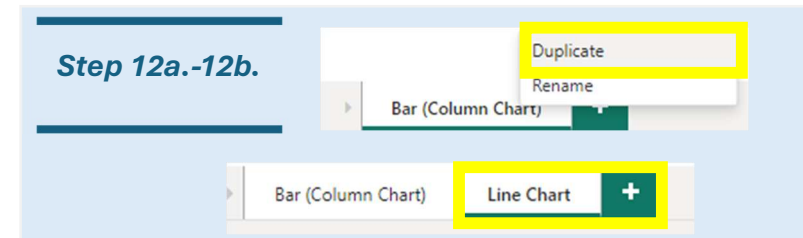


11. Let's do the simplest thing and make it clearer what the visualization shows by giving a subtitle
- a. Under the “Title” turn on “Subtitle” option and add “*Data presented includes Q1-Q3 for 2024”



Line Chart

12. Let's make a line chart. We could start from scratch, or we can **edit the one we just made...** let's do that
- a. On the bottom of the page, right click and then “**Duplicate**” and then with the duplicated tab rename to “**Line Chart**”
- b. Then select the existing bar chart and in the visualization, pane select the “**Line Chart**” icon
- c. Change the value in the **X-axis** shelf to “**Year**” and add a legend with “**Loan Type**” (keep Y-axis as “Sum of Dollars Outstanding”)



d. Turn the **Markers** option **on** under **Visual** setting

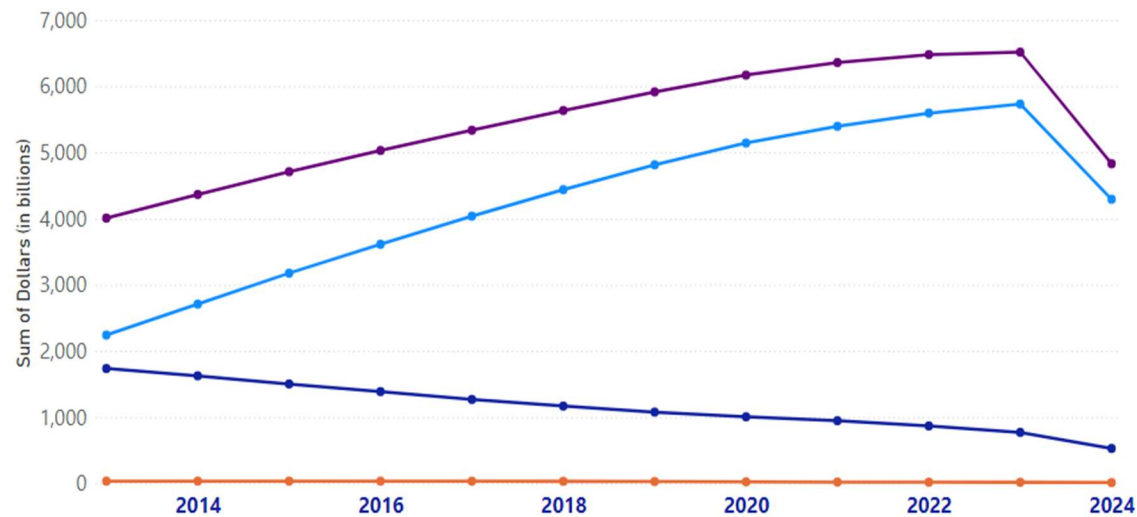


Final Expected Visualization

US National Outstanding Loan Amounts 2013-2024* by Type

*Data presented includes Q1-Q3 for 2024

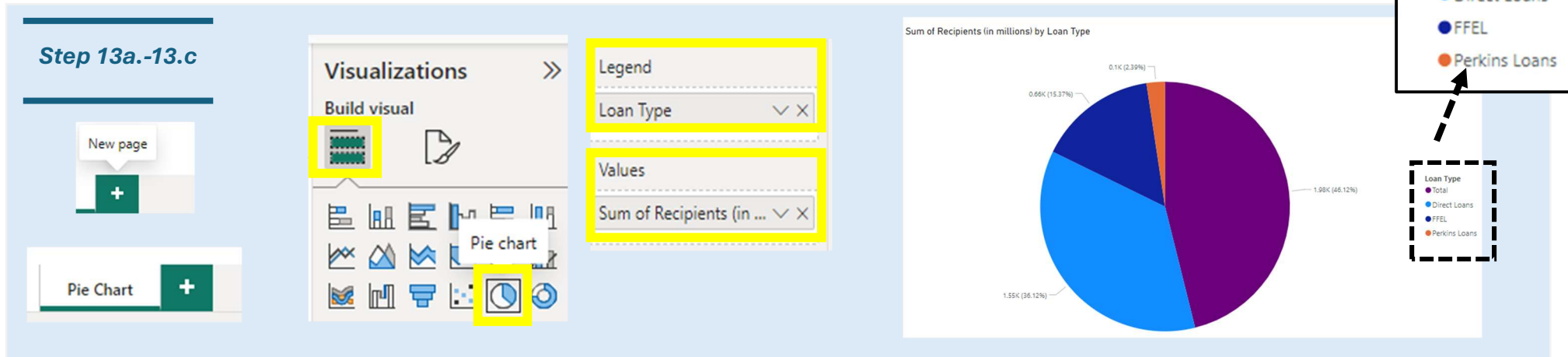
Loan Type ● Direct Loans ● FFEL ● Perkins Loans ● Total



Pie Chart

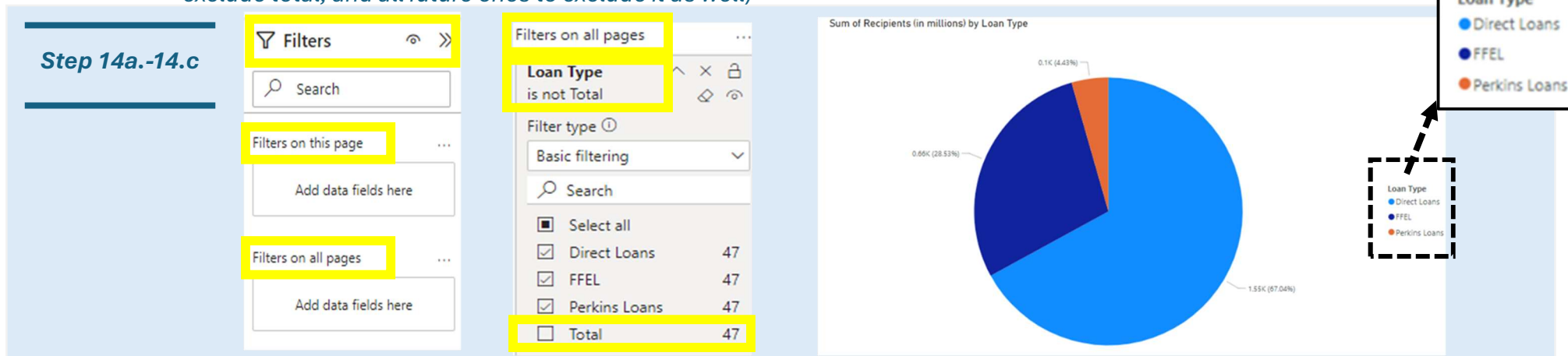
13. Let's make a pie chart.

- Create a new page and rename it to **"Pie Chart"**
- Select the **"Pie Chart"** icon
- Add **"Loan Type"** to the legend shelf and **"Recipients (in millions)"** on the values shelf



14. Let's use filtering to **remove Total** from the pie chart since this breaks the **Parts of Whole** rule (sum of the pie should be 100%)!

- Power BI has two options for filtering visualizations: **1) Filter by / on a single page** or **2) Filter on all pages**
- Under the **"Filters"** options add **"Loan Type"** under the **"Filters on this page"** and unselect **"Total"**
- Now move **"Loan Type"** to also test out **"Filters on all pages"** and see how they impact the pie chart, and then the other visuals: Bar and Line charts. **Keep the filter on "Filter on all pages"** (this will change all charts we already made to now exclude total, and all future ones to exclude it as well)

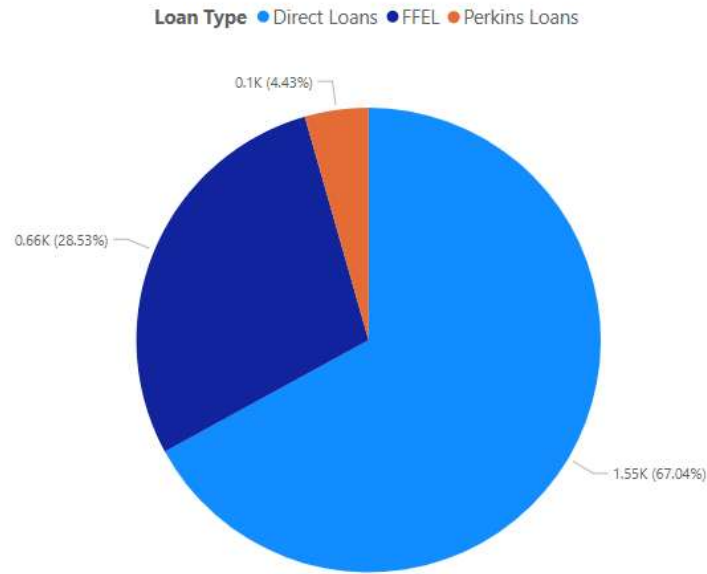


- d. Now looking at the visual shown below make edits to make your visualization to the one shown
 - i. Add the following Title to the visual **“Total US National Loan Recipients 2013-2024* by Type”**
 - ii. Title Font size: **25**, the Title Font color and Weight: **“#12239E, Theme color 2”** and **Bold** the text
 - iii. Add subtitle **“*Data presented includes Q1-Q3 for 2024”**
 - iv. Change legend’s position to **“Top center”**, increase the text size to 12

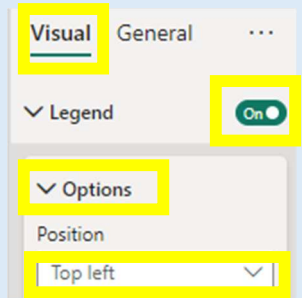
Final Expected Visualization

Total US National Loan Recipients 2013-2024* by Type

*Data presented includes Q1-Q3 for 2024



Step 14d.



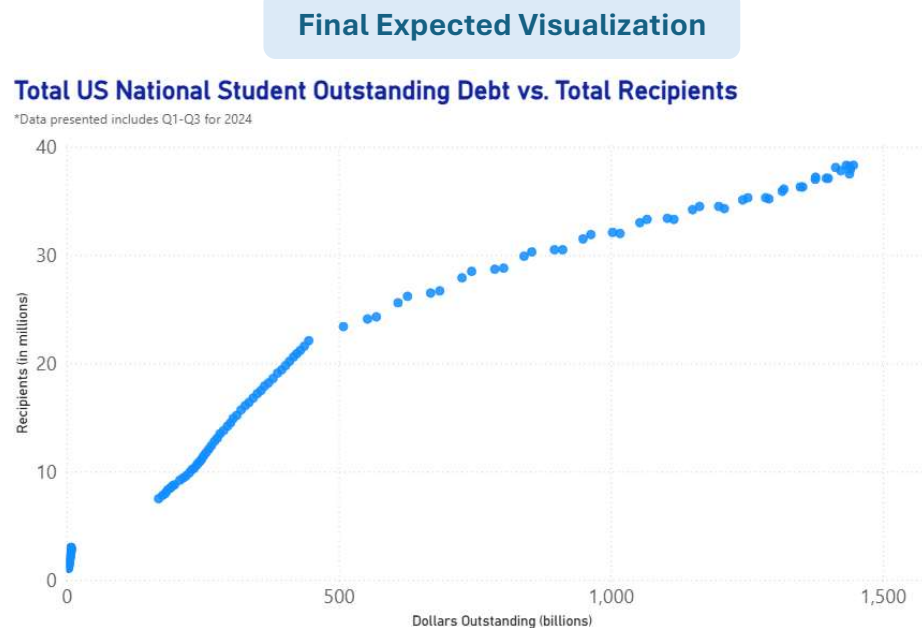
Scatter Plot

15. Let's create a scatterplot
 - a. Create a new page and rename it to **“Scatter Plot”**
 - b. Select the **“Scatter chart”** icon
 - c. Add **“Dollars Outstanding”** to the **X-axis** and **“Recipients”** to the **Y-axis**
 - d. Right click on the **“Sum of Dollars Outstanding”** under the **X-axis shelf** and change to **“Don't Summarize”**
 - e. Repeat for step d. for values on the **Y-axis**

Step 15a.-15.e



- f. Now looking at the visual shown below make edits to make your visualization match the general style of the one shown below
- Add title **“Total US National Student Outstanding Debt vs. Total Recipients”**
 - Title Font size: **25** and Title Font color / Weight: **“#12239E, Theme color 2”** and **Bold the text**
 - Change **x and y axis title size to 15**
 - Add subtitle **“*Data presented includes Q1-Q3 for 2024”**



Note: if your chart doesn't look like this and there are extra dots above the 1,000 to 1,500 range this means you didn't apply the filter from Pie Chart to ALL PAGES, and instead to a single page or single visual.

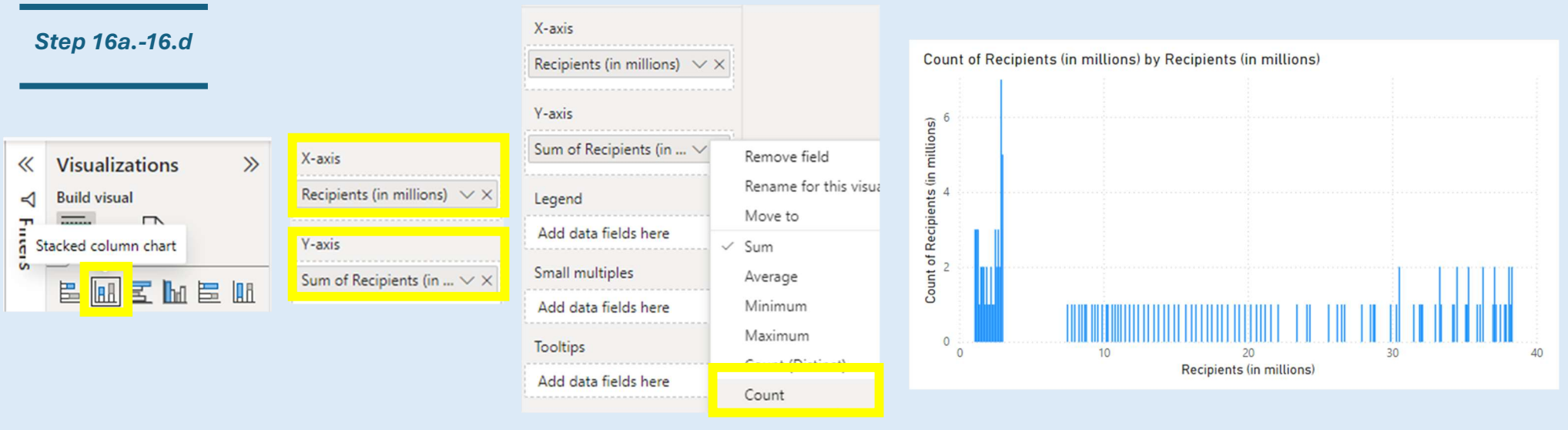
Histogram

16. Let's create a histogram

- Create a new page and rename it to **"Histogram"**
- Select the **"Stacked column chart"** icon (we could also use the other clustered(grouped) column charts as well)
- Add **"Recipients (in millions)"** to **X-axis** and also to the **Y-axis**
- Change value on y-axis to **"count"** by right clicking on the arrow and selecting **"Count"**

Note: it won't look like a histogram just yet, we need to bin the data in step 17.

Step 16a.-16.d



17. Now let's bin the data to make it look more like a classic histogram. To do this in Power BI we will need to **create a grouping** of the **y-axis** feature **"Recipients (in millions)"** to mimic **bins** for the **X-axis**

- On the **right-hand side** where the data column **"Recipients"** is listed **right click on the three dots** and select **"New group"**. Then name the group **"Bins Recipients (in millions)"**
- Under **"Bin type"** select change to **Number of bins** and set bin count to **5**, then select **"OK"**
- Now replace the **current value** on the **X-axis** with this new group feature **"Bins Recipients (in millions)"**
- Now to make the visualization look more like a histogram, let's remove the spaces between the bars, under the **Format Visual**, under **Columns** and **Layout** option change **"Spaces between categories"** to **0%**.

Step 17a.-17b.

☐ Σ Recipients (i...
Collapse all
Expand all
New group
Add to filters
Add to drill through

Groups

Name *

Bins Recipients (in millions)

Field

Recipients (in millions)

Group type

Bin

Bin type

Size of bins

Min value

1.1

Size of bins

Number of bins

Binning splits numeric or date/time data by an amount you specify.

Bin count *

5

Step 17c.-17d

Visual

Columns

X-axis

Recipients (in millions) X

X-axis

Bins Recipients (in mill... X

Layout

Reverse order

Off

Sort by value

Off

Space between categories

0 %

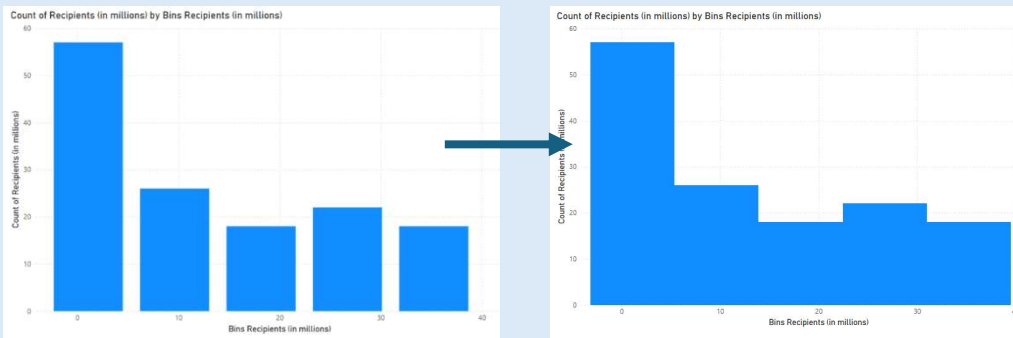
Space between series

0 px

- e. Now let's add the final touches to the data visualization to match the one shown
- Add title **"Distribution of Total US National Loan Recipients 2013-2024"**
 - Title Font size: **25** and Title Font color / Weight: **"#12239E, Theme color 2"** and **Bold the text**
 - Change **x-axis title** to **"Recipients (in millions)"**, **y axis title** to **"Count"** and make both titles size **15**
 - Add subtitle **"*Data presented includes Q1-Q3 for 2024"**

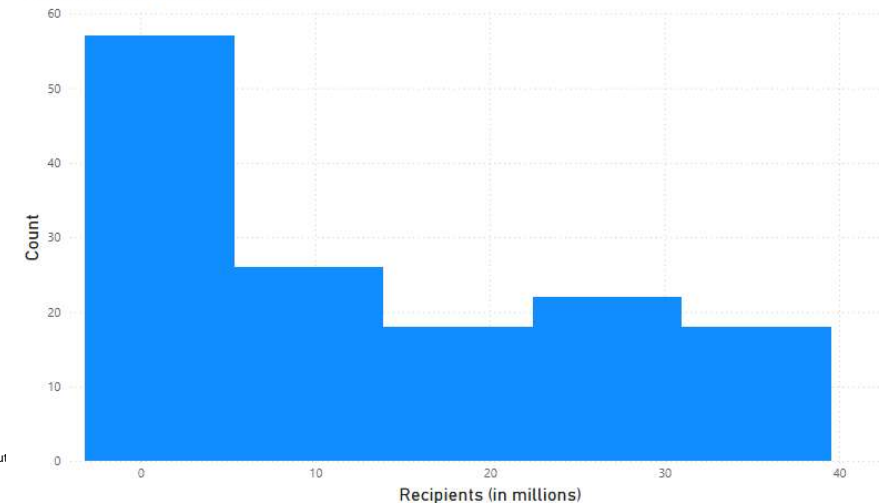
Final Expected Visualization

Before and After 17 d.



Distribution of Total US National Loan Recipients 2013-2024*

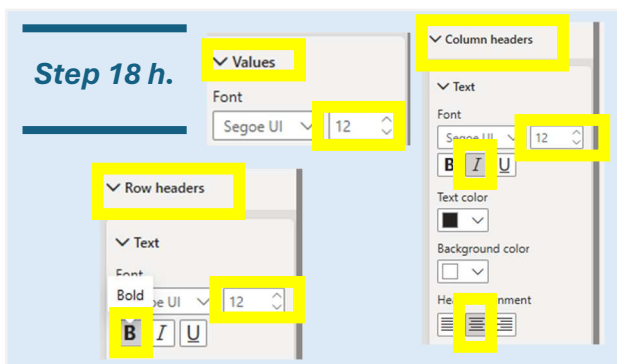
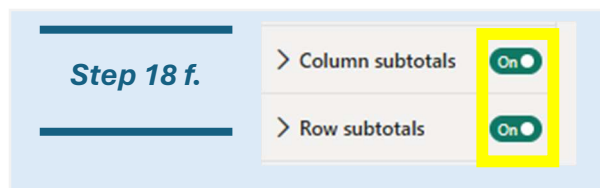
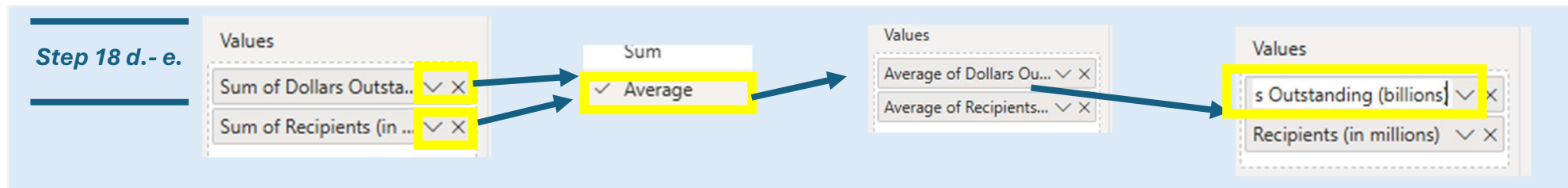
*Data presented includes Q1-Q3 for 2024



Table

18. Let's create a table showing the

- Create a new page and rename it to **"Table"**
- Select the **"Matrix Table"** icon (we could also use the other table icon as well)
- Add **"Loan Type"** to **Rows** and add **"Dollars Outstanding (billion)"** and **"Recipients (in millions)"** to **Values**
- Change from **"Sum"** to **"Average"** for both metrics in c.
- Double click into each of the metrics in the **Values** shelf, and change their titles to **"Dollars Outstanding (billions)"** and **"Recipients (millions)"** respectively. (i.e. removing "Average of")
- Under "Format Visual", **turn off** the **Column subtotals** and **Row Subtotals**
- Under "Format Visual", change the **text size for Values to 12**, and change **Row headers** font size 12, **bolded** and **Column headers font size 12**, center alignment and *italicize* the text
- Add a title **"Average Dollars Outstanding and Recipients by Loan Type"**, Title Font size: 15 and Title Font color / Weight: **"#12239E, Theme color 2"** and **Bold** the text



Final Expected Visualization

Average Dollars Outstanding and Recipients by Loan Type

Loan Type	Dollars Outstanding (billions)	Recipients (in millions)
Direct Loans	1,088.91	32.94
FFEL	295.06	14.02
Perkins Loans	6.37	2.18